

Ablation summary: selecting a balanced PVI-GCNM

Deployable homogeneous-baseline models; stage 2 on the same 32 exact-nonlinear holdout phantoms. One factor changes at a time.

Experiment	Global corr. ↑	Dice ↑	RMSE ↓	BG RMS ↓
Direct core	0.458	0.230	0.00484	0.000581
Direct + coordinates	0.561	0.471	0.00444	0.000865
$\alpha = 1$	0.557	0.543	0.00430	0.001541
$\alpha = 2$	0.524	0.457	0.00445	0.001778
$\alpha = 4$	0.489	0.407	0.00477	0.002232
$\alpha = 8$	0.479	0.394	0.00519	0.002730
$\alpha = 1, \beta = 0.25$ * selected	0.557	0.485	0.00432	0.001276
$\alpha = 1, \beta = 0.5$	0.530	0.462	0.00446	0.000992
$\alpha = 1, \beta = 1.0$	0.527	0.427	0.00453	0.000845

Takeaway. Coordinates give the largest single gain (0.458→0.561). $\alpha = 1$ best balances Dice and reconstruction error; $\beta = 0.25$ trims background at little correlation cost. The model was selected by a composite RMSE/background/Dice score, not by maximizing one column.

Bold = best value in each metric column; * = selected operating point. The selected row need not win every individual metric.

Source: technical report Table 10; selected checkpoint faithful_background025_hom_seed0.